

Omkar Singh

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SUMMARY

Full-stack developer and B.Tech student at IIT Kharagpur with hands-on experience building distributed systems in Go and production-grade web applications with the MERN stack and React.js. Designed backends processing 50–100 orders/sec at sub-ms latency. Open to SDE internship; shipped 3 live projects handling real concurrency and distributed state.

EDUCATION

Indian Institute of Technology (IIT) Kharagpur

B.Tech – Industrial & Systems Engineering

2023 – Present

GPA: 7.56 / 10.0

EXPERIENCE

JPMorganChase – Software Engineering Externship

Apr – May 2026

Software Engineering Virtual Experience

Remote

- Built a Kafka-backed async transaction pipeline in Spring Boot — consumed, validated, and persisted high-volume messages to H2 via JPA, reducing processing coupling across 3 relational tables.
- Implemented transaction validation and persistence with Spring Data JPA and H2 SQL, modelling entities and balance updates across 3+ relational user-record tables.
- Developed a REST endpoint for user-balance queries via a Spring controller; connected the service to an external Incentive API using `RestTemplate`, incorporating responses into end-to-end transactional workflows.
- Validated end-to-end system reliability across message ingestion, DB persistence, and external API integration by authoring 50+ JUnit test cases in Maven; achieved zero regression failures across all transaction workflows.

PROJECTS

Synthetic-Bull | *Go, Gin, GORM, MySQL, WebSockets, REST APIs, Redis* | [Live](#)

2025

- Architected a horizontally scalable 11-node distributed Go backend handling real-time order matching, portfolio management, and market data aggregation across decoupled microservices; applied system design principles to achieve sub-ms inter-service latency.
- Implemented a custom Limit Order Book with Price-Time Priority: $O(\log N)$ insertion via Red-Black Trees, $O(1)$ best-price retrieval, supporting partial fills across concurrent sessions.
- Engineered lock-free fan-out via goroutines and `sync.WaitGroup` broadcasting to 5 services at sub-ms latency; integrated a Redis low-latency caching layer for market state; exposed REST API Gateway + WebSocket endpoints for tick-by-tick candlestick rendering.
- Built a GBM market simulator generating 50–100 realistic orders/sec, driving live OHLCV candlestick aggregation with real-time frontend updates.

Collaborative Whiteboard | *React.js, Firebase Firestore, Canvas API, Tailwind CSS* | [Live](#)

Aug 2025

- Architected a multi-user drawing app supporting 10+ concurrent users with sub-100ms sync latency via Firebase Firestore real-time listeners propagating canvas state across all clients.
- Built a stateful drawing engine with 6 tools and full undo/redo history via Canvas API and Rough.js; designed a mobile-first responsive UI with React Hooks.

MERN E-Commerce Platform | *MongoDB, Express.js, React.js, Node.js, JWT, Multer* | [Live](#)

2025

- Built a full-stack e-commerce platform with JWT auth via HTTP-only cookies (XSS-safe), role-based admin controls, product image upload via Multer, and complete order + review management.
- Designed a RESTful API with Express.js and Mongoose featuring async error-handling, bcrypt hashing, and protected route guards; deployed React SPA to Netlify with a concurrent dev workflow.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript (ES6+), TypeScript, SQL
Frameworks & Libs	React.js, Node.js, Express.js, Spring Boot, Gin/GORM, Tailwind CSS
Databases	MongoDB, MySQL, Firebase Firestore, H2
Tools & Platforms	Git, Docker, Kafka, JWT, Maven, Postman, AWS, Redis
Concepts	Microservices, RESTful APIs, WebSockets, Full-Stack Development, DSA, Agile/Scrum, OOP, System Design

CERTIFICATIONS

Introduction to Software Engineering Job Simulation | Commonwealth Bank (Forage)

May 2026

Software Engineering Job Simulation | Quantum

May 2026

Advanced Software Engineering Simulation | Walmart Global Tech (Forage)

Sep 2025

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

LeetCode	Solved 250+ problems across arrays, trees, graphs, and dynamic programming
Codeforces	Rating 1286 (Specialist) – consistent performance in algorithmic contests
HackerRank	5-Star Gold Badge in Problem Solving